

Software Requirements Specification

Titans InstaFood — Requirements Specification

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Revision history

Version	Date	Author	Summary
1.0	2024-02-09	CIEM IT CONSULT - SO	Original PDF baseline
2.0	2026-05-23	Requirements package	Acceptance criteria; FR15 retired; NFR browser split
3.0	2026-05-23	Build spec	OpenAPI, RBAC, state machines, UI/integration depth
3.1	2026-05-23	Deliverable fixes	Cost/test workbook corrections
3.2	2026-05-23	Package layout	Professional file names; deliverables/ structure
4.0	2026-05-24	Production baseline	Full SDLC documentation audit fixes; traceability and API alignment

1. Functional requirements — Release 1 (MVP)

Priority: Must unless marked *Should*.

1.1 Customer profile and identity

ID	Requirement	Acceptance criteria
FR1	Customers can create a profile with essential information.	Given valid required fields, account is created and confirmation is sent if enabled.
FR2	Users can update profile details (contact, preferences).	Given an authenticated user, profile changes persist and display on next load.
FR3	Users can delete their account; data is permanently removed.	Given deletion confirmed, account is disabled; PII purged/anonymized per DRRD5 and other applicable regulations.
FR4	The system associates users with order history for personalized history, recommendations or empty-state display from history.	Given valid history, recommendations or empty-state display from history.

1.2 Menu and catalog (admin)

ID	Requirement	Acceptance criteria
FR5	Administrators can add/update menu items (name, description, price), CRUD persists; item visible on customer menu.	Given valid admin role, CRUD persists; item visible on customer menu.
FR6	Administrators can mark items unavailable or remove items.	Given unavailable flag, item is hidden or not addable by customers.
FR7	Administrators can associate offers/discounts with menu items.	Given valid promotion linked to item, discount reflected at item/checkout.

1.3 Cart and ordering

ID	Requirement	Acceptance criteria
FR8	Customers select menu items, add to cart, specify quantity.	Cart line shows item name, quantity, and correct subtotal.
FR9	Users update cart before confirming order.	Quantity and removals persist until checkout or session expiry.
FR10	Customers view cart items at all times while shopping.	Cart indicator accessible from all shopping pages.
FR11	Customers save cart for later purchase.	Given saved cart, when user returns, cart is restored.
FR12	Customers delete items from cart.	Removing last item shows empty cart state.
FR23	Customers choose pickup option at checkout.	Checkout offers pickup; order record stores FulfillmentType = PICKUP.
FR24	System shows full restaurant menu by category, 24/7.	Categories visible: Breakfast, Lunch, Dinner, Drinks; menu loads at all hours.

1.4 Payments

ID	Requirement	Acceptance criteria
FR13	Secure payment gateway integration for online transactions.	TLS 1.2+; card data not stored in clear text; gateway token only.
FR14	Customers can update stored payment methods.	Given authenticated customer, only token/last-four shown after gateway update.
FR15	(retired) Duplicate of FR14.	Retired v2.0 — not implemented.
FR16	Payments associate with specific orders.	Each Payment row references exactly one OrderID.

1.5 Order tracking and discovery

ID	Requirement	Acceptance criteria
FR17	Customers query order status in real time.	Status updates within 30 seconds of backend change.
FR18	Customers review past order history.	Paginated list, newest first; includes date, items, total, status.
FR19	Customers search menu items quickly.	Search returns relevant items within 3 seconds (p95, 500 items).
FR20	Customers filter by price and dietary preferences.	Filters combine with AND logic; filtered list matches criteria.

1.6 Reporting and promotions (admin)

ID	Requirement	Acceptance criteria
FR21	Generate daily, weekly, monthly sales reports.	Reports export or view with date range; totals match order data.
FR22	Insights on popular items and preferences.	Report lists top N items by quantity/revenue for period.
FR25	Reports on payment transactions linked to orders.	Each row includes payment id, order id, amount, status.
FR26	Administrators manage promotional offers.	CRUD on promotions with start/end dates and rules.
FR27	Administrators delete/deactivate promotions.	Deactivated promotions not applied at checkout after end.
FR28	Reports on feedback, ratings, reviews.	Aggregate rating and review count per period.
FR29	Reports on effectiveness of offers/discounts.	Compare order volume/revenue with and without promotion.
FR30	Administrators respond to reviews.	Public response visible on review thread; audit metadata stored.

2. Functional requirements — Release 2 (R2)

ID	Requirement	Acceptance criteria
FR31	Guest checkout without account	Order completes with email/phone only; no password required.
FR32	Order confirmation email	Email sent within 5 minutes of successful payment.
FR33	Coupon codes at checkout	Valid code reduces total; invalid/expired rejected with message.
FR34	Special instructions on order	Free-text stored on order; visible to kitchen/admin.
FR35	Estimated delivery/pickup times at checkout	ETA displayed before payment.
FR36	Real-time delivery tracking	Map or status timeline for delivery orders.
FR37	Post-order rating and review	One review per completed order per customer.
FR38	One-click reorder from history	Prior items added to cart in one action.
FR39	Search within order history	Filter past orders by item name or date.
FR40	Edit delivery address before place order	Address validated; optional save to profile.
FR41	Forgot password	Reset link expires within 24 hours.
FR42	Password complexity on registration	Min 8 chars with upper, lower, digit, and special.
FR43	Nutritional information on request	Calories/ingredients from catalog metadata.
FR44	Multiple payment methods	Card plus at least one alternative per gateway.
FR45	Notifications (confirm, status, delivery)	Per user preference; channel defined at R2 design.
FR46	Cross-device intuitive UI	Usability test pass rate $\geq 80\%$ on task script.
FR47	Order minimums and delivery zones	Orders below minimum or outside zone blocked with message.
FR48	Secure login (CAPTCHA, etc.)	Rate limiting and CAPTCHA after N failed attempts.
FR49	Customer support contact	In-app link/form; ticket or mailto logged.
FR50	Audit log of user actions and system events	Immutable append-only log with user and timestamp.

3. Authentication (supplementary)

ID	Requirement	Acceptance criteria
AUTH-01	Registered users log in with username/email and password.	Valid credentials issue JWT; invalid credentials rejected without disclosure.

4. Non-functional requirements

ID	Requirement	Acceptance criteria
NFR1	Two-factor authentication for accounts	TOTP or SMS 2FA optional for customers; required for admin.
NFR2	Audit logs for logins, data changes, orders	All security events logged with timestamp, user, action.
NFR3	Encrypt payment data in transit and at rest	TLS 1.2+; DB encryption or token-only card storage.
NFR4	Scale users/orders without degradation	Load test: 100 concurrent users, p95 API reads < 500ms.
NFR5	24/7/365 availability	99.5% monthly uptime excluding planned maintenance.
NFR6	Backup and recovery	RPO <= 24h, RTO <= 4h documented and tested annually.
NFR7	Desktop, tablet, smartphone	Responsive layouts 320px–1920px without horizontal scroll.
NFR8	User-friendly UI (customer + admin)	SUS score >= 68 on pilot (*Should*).
NFR9	Clear navigation and responsive design	Primary tasks <= 3 clicks from home.
NFR10	Payment gateway and third-party integration	Webhook retry and idempotency documented and verified.
NFR11	Scalable database	Vertical scale path documented; connection pooling enabled.
NFR12	Fast order placement and browsing	p95 place-order < 2s; p95 menu load < 1.5s.
NFR13	Admin dashboard updates within 10s	WebSocket or polling interval <= 10 seconds.
NFR14	Chrome browser compatibility	Last two major versions pass functional smoke tests.
NFR15	Edge browser compatibility	Last two major versions pass functional smoke tests.
NFR16	Firefox browser compatibility	Last two major versions pass functional smoke tests.
NFR17	Safari browser compatibility	Last two major versions pass functional smoke tests.
NFR18	Menu search performance	p95 <= 3 seconds with 500 menu items.
NFR19	iOS 15–17 support	Mobile web smoke tests pass on supported iOS versions.
NFR20	Accessibility	WCAG 2.1 Level AA for customer flows; no critical violations.
NFR21	Failover to backup systems	DR drill: switch to standby <= 60 minutes annually.

5. Traceability

- SDLC/03-Requirements-Analysis/RTM-Requirements-Traceability-Matrix-v4.0.xlsx
- SDLC/05-Testing/STP-System-Test-Plan-v4.0.xlsx
- SDLC/03-Requirements-Analysis/DRD-Data-Requirements-Document-v4.0.pdf
- SDLC/04-Design/DDD-Logical-Data-Dictionary-v4.0.pdf
- SDLC/03-Requirements-Analysis/UCS-Use-Case-Specification-v4.0.pdf
- SDLC/04-Design/API-REST-Contract-Specification-v4.0.pdf